

Before Optimization: A Short Introduction to the Survival-Breakdown Model

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Note on the English Version

The authoritative original version of this paper is the Japanese-language original.

This English version was prepared with AI-assisted translation support because the author is a native speaker of Japanese and not a native speaker of English.

This English version is provided for reference only. It should be treated as supplementary material accompanying the original Japanese paper. In the event of any discrepancy in wording, nuance, or interpretation, the Japanese original shall prevail.

Abstract

This paper provides a short English-language introduction to the Survival-Breakdown Model (SBM). Its central claim is that economic theory should not begin from optimization, equilibrium, or aggregate consistency taken by themselves. It should begin from more prior conditions that determine whether agents and organizations are constituted as operative units, whether they remain survivable, whether they retain a basically feasible range of action, and whether continuation through time is maintained. In SBM, the explanatory order is: constitutive conditions, survivability, basic feasibility, continuation, and only then optimization.

On that basis, the paper argues that demand-supply theory, general equilibrium theory, and macroeconomics should not be treated as universal first principles. They remain analytically useful, but only within domains in which prior conditions have already been secured. In SBM terms, they are not abolished but reclassified as lower-order and conditionally valid theories of local, systemic, and aggregate adjustment. The paper also clarifies what SBM does not claim: it does not deny that standard theory can formally represent hardship or collapse-like states in some ways, it does not merely add one more severe constraint to optimization, and it does not present itself as a fully completed closed system. Rather, SBM should be understood as a theory of prior conditions and as a research program with an established core and an open frontier.

Keywords

Survival-Breakdown Model; economic agency; theoretical order; survivability; basic feasibility; continuation; standard theory; general equilibrium; macroeconomics

JEL Classification

B41, B49, D01, D11, D50, E10

1. Introduction

Modern economics usually begins from agents who maximize objective functions under constraints. Households maximize utility, firms maximize profit, and from there demand, supply, prices, equilibrium, and aggregate outcomes are derived. SBM does not deny the analytical power of this construction. Its claim is narrower but more foundational: this does not yet show that optimization is the correct starting point of theory. If agents and organizations can enter states of non-survival, breakdown, inability to continue, or inability to participate, then before asking how they optimize, one must ask whether they are constituted as operative agents at all and whether they possess the minimum conditions required for meaningful action. SBM therefore proposes a reordering of economic theory's starting point.

The issue is not simply whether survival or breakdown can be represented mathematically somewhere within standard theory. It is whether such representations identify the correct explanatory starting point. Even if threshold-like states can be written through constraints, penalties, or state-dependent utilities, representability does not settle theoretical priority. What SBM contests is not the local usefulness of standard models, but their claim to first-principle status.

This paper is an introductory route into the broader SBM program. It does not restate the whole architecture of the three longer papers. Its aim is to present the core claim in compact form, show a minimal formal sketch, explain why standard theory becomes lower-order, and indicate how the three existing SBM papers fit together as a connected program.

2. The Core Claim of SBM

The core claim of SBM is that economics should begin from prior conditions of agency rather than from optimization, equilibrium, or aggregate consistency. The order proposed by SBM is:

constitutive conditions -> survivability -> basic feasibility -> continuation -> optimization

This is not a pedagogical sequence but a claim about explanatory rank. Constitutive conditions provide the material basis of agency. Survivability asks whether the relevant unit remains above the threshold below which continuation is impossible. Basic feasibility specifies the room for meaningful action remaining after survival is secured. Continuation asks whether that agency is maintained, destabilized, or lost through time. Optimization appears only after these prior conditions have been established.

This reorders the concept of the economic agent. In standard theory, the agent is generally introduced as an already given chooser. SBM argues that this is explanatorily too late. Before one can speak of choice, one must ask whether the relevant unit is survivable and continuable as an agent at all.

For that reason, SBM should not be understood as a standard optimization model with one more severe boundary condition. A constraint limits the behavior of an already constituted agent within an already given

domain. SBM asks whether that agent and domain are constituted in the first place. It is therefore better understood as a theory of prior conditions than as an added-constraint model.

3. A Minimal Formal Sketch

A minimal formal skeleton of SBM is enough to show why it is not merely rhetorical. Let R denote the resources available to an agent and L the minimum level required for continued existence or operation. Then the survival condition is

$$R \geq L \quad (3.1)$$

and the breakdown condition is

$$R < L \quad (3.2)$$

The theoretical point is that this threshold is prior to optimization. If the minimum condition for continuation is not satisfied, the first issue is not which option is optimal but whether agency can continue at all.

Survival alone is still not sufficient for meaningful action. SBM therefore defines basic feasibility as the slack remaining above survival:

$$B = R - L \quad (3.3)$$

If B is small or negative, the domain of meaningful action is narrowed or lost. This is why the feasible action set is not primitive in SBM; it depends on prior feasibility. Theoretical order can therefore be expressed compactly as

$$\text{survivability} > \text{basic feasibility} > \text{optimization} \quad (3.4)$$

Optimization is not abolished, but it is positioned as a posterior operation.

To represent continuation dynamically, let B_t denote continuation balance at time t , u_t denote inflows, and $c > 0$ denote minimum continuation cost. Then:

$$B_{t+1} = B_t + u_t - c \quad (3.5)$$

Given a destabilization threshold $\theta > 0$, the agent's state can be classified as:

$$B_t > \theta \Rightarrow \text{continuation} \quad (3.6)$$

$$0 < B_t \leq \theta \Rightarrow \text{destabilization} \quad (3.7)$$

$$B_t \leq 0 \Rightarrow \text{termination / functional breakdown} \quad (3.8)$$

This expresses one of SBM's main departures from standard theory: agency is not treated as stably given, but as something maintained, eroded, or lost through time.

4. Why Standard Theory Becomes Lower-Order

If agency must first be understood through constitution, survivability, feasibility, and continuation, then theories that begin from already operative agents cannot be first principles. This applies to demand-supply theory, general equilibrium theory, and macroeconomics. SBM does not abolish them. It reclassifies them as lower-order theories whose validity depends on prior conditions already being secured.

Under SBM, demand-supply theory becomes a theory of local market adjustment. General equilibrium becomes a theory of systemic adjustment over an operative participant domain. Macroeconomics becomes a theory of aggregate adjustment where the continuation structure of the economy remains sufficiently intact. All three remain analytically useful, but none any longer counts as the first beginning of economics. Their common status is that of conditional subordinate theories resting on a more prior layer.

That is why the right formula is not abolition but reclassification. Standard theory is preserved where its conditions of application are met. But its universal foundational status is lost. The issue is not utility versus uselessness, but explanatory rank.

5. What SBM Does Not Claim

Three clarifications are sufficient for an introductory version.

First, SBM does not claim that standard theory is wholly incapable of representing survival-like conditions, severe distress, or breakdown-like states. It accepts that such states can often be represented in some fashion. But this establishes only representability, not first-principle status. The issue is what must come first in explanation.

Second, SBM does not claim merely to add one more severe constraint to optimization. That interpretation notices only a superficial formal resemblance and misses the difference in theoretical level. SBM asks whether the agent, the feasible domain, and the participant set are constituted at all. It therefore resets the domain on which later theory operates.

Third, SBM does not claim to be a fully completed closed replacement system. It has already established a core thesis and a basic architecture, but many lower-order developments remain open. It is more accurate to understand SBM as a research program with an established nucleus and further tasks ahead.

6. The Three-Paper Structure

The current SBM program is best understood through its three existing papers.

The first paper establishes the foundational order of economic agency: constitution, survivability, basic feasibility, continuation, and then optimization.

The second paper takes that order as given and reclassifies demand-supply theory, general equilibrium theory, and macroeconomics as lower-order special cases.

The third paper organizes the main criticisms, distinguishes representability from first-principle status, and clarifies how SBM should be positioned relative to standard theory.

This yields a determinate sequence:

foundation -> reclassification -> clarification

The three papers should therefore be read not as isolated interventions but as one connected research program. The present paper is only an entrance to that sequence.

7. Conclusion

The central conclusion of SBM can be stated simply. Economics should begin not from optimization alone, but from the prior conditions that make agency, participation, and continuation possible. Standard theory remains valid where those prior conditions are stably secured, but it no longer stands as the universal first principle of economics. SBM is therefore best understood as a foundational reordering of theoretical levels and as a research program for reconstructing the domain and sequence of economic explanation on that basis.

References

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